

Random Forest Based Wine Quality Classification

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Introduction

Machine learning techniques are widely used for prediction and classification tasks by identifying patterns within datasets. This project focuses on applying the Random Forest classification algorithm to predict wine quality using physicochemical properties from the WineQT dataset. Wine quality prediction is a useful example of supervised learning because the quality of wine is influenced by several measurable chemical features such as acidity, alcohol content, sulphates, density, pH, and sugar levels. The objective of this study is to analyze these features, train a Random Forest model, evaluate its classification performance, and understand the influence of different model parameters such as the number of decision trees and maximum tree depth. The report presents dataset analysis, model methodology, experimental results, visualizations, and feature importance evaluation to understand the effectiveness of Random Forest for wine quality classification.

Random Forest Based Wine Quality Classification

This tutorial demonstrates how Random Forest can be used to classify wine quality using chemical properties from the WineQT dataset. The main focus is understanding how Random Forest parameters such as number of trees and tree depth influence prediction performance.

Dataset Description

The dataset contains 1143 wine samples with physicochemical features including acidity, sugar content, chlorides, sulphates, density, pH and alcohol. There were no missing values. The original quality scores were converted into a binary classification problem: low quality wine and high quality wine.

Model Methodology

Random Forest is an ensemble learning algorithm that combines multiple decision trees. Each tree learns from different samples of the data, and the final prediction is created by combining the decisions of many trees. This reduces overfitting compared with a single decision tree.

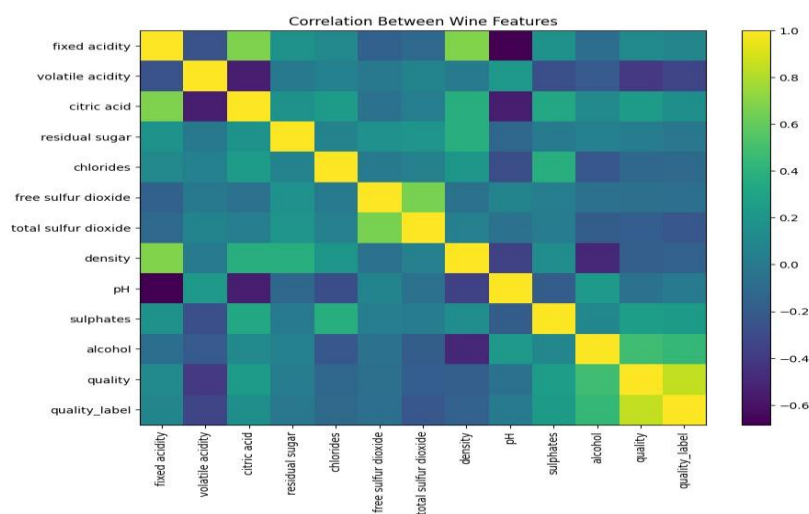
Model Results

The optimized Random Forest model achieved an accuracy of approximately 75.98%. The weighted F1-score was 0.76, showing balanced performance between low-quality and high-quality wine classes. The confusion matrix showed 174 correct predictions from 229 test samples

A bar chart titled "Distribution of Original Wine Quality Scores". The y-axis is labeled "Number of Samples" and ranges from 0 to 500 in increments of 100. The x-axis is labeled "Wine Quality Score" and ranges from 3 to 8. The chart shows the frequency of each quality score. The highest frequency is for a score of 5, with approximately 485 samples. Scores 6 and 7 also have high frequencies, around 465 and 145 respectively. Scores 3, 4, and 8 have much lower frequencies, around 10, 35, and 20 samples respectively.

Wine Quality Score	Number of Samples
3	10
4	35
5	485
6	465
7	145
8	20

Figure 2: Correlation Heatmap - displays relationships between chemical properties and wine quality:



The correlation heatmap helps identify relationships between different chemical properties of wine. Stronger relationships between features can influence the prediction process. These correlations provide useful insights before training the Random Forest model.

Figure 3: Number of Trees Experiment - compares Random Forest accuracy when changing ensemble size:

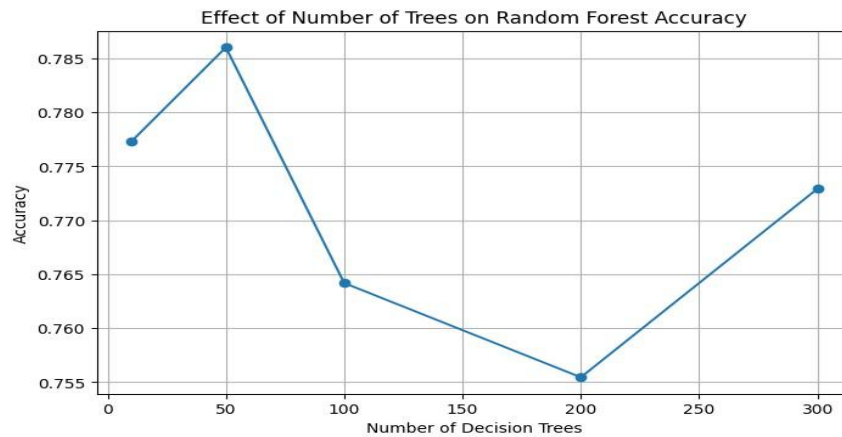
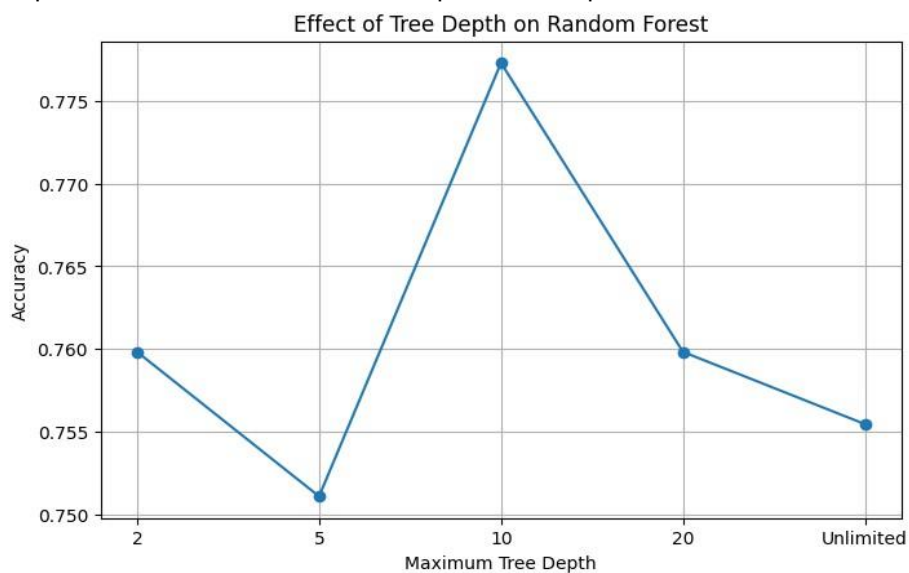
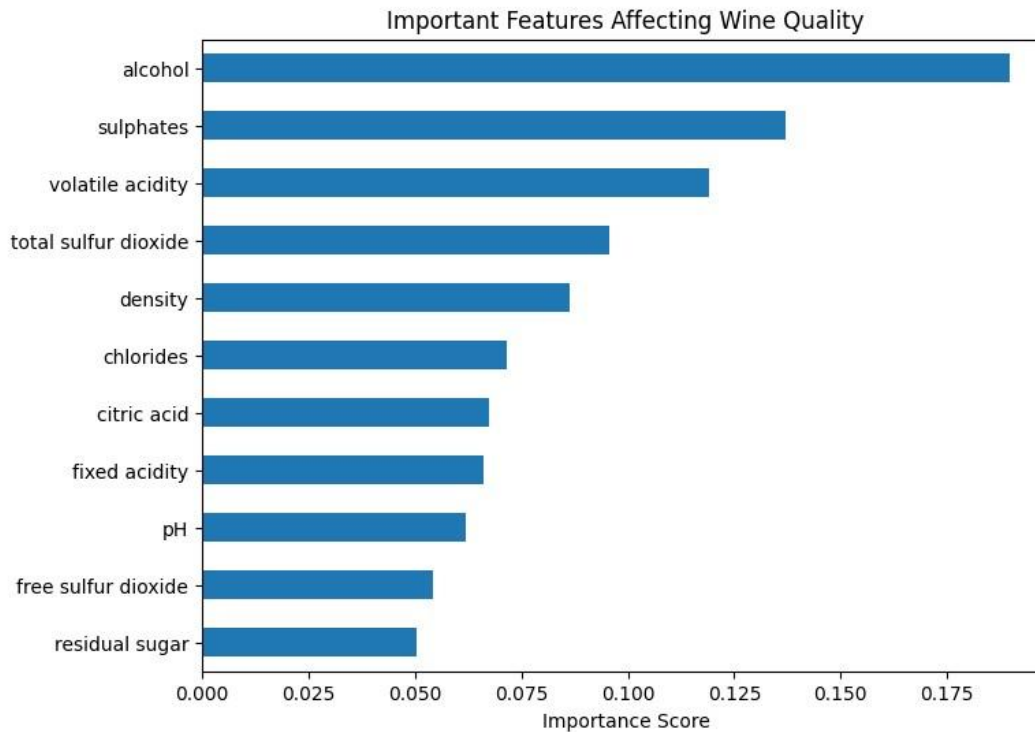


Figure 4: Tree Depth Experiment - demonstrates the influence of model complexity:

The number of trees experiment shows how changing the ensemble size affects accuracy. Increasing trees can improve stability, but very large values do not always guarantee better performance. The experiment helps select suitable Random Forest parameters. p



5: Feature Importance - identifies the most influential wine properties: The tree depth and feature importance analysis explains model behaviour. Tree depth controls complexity.



Conclusion

The experiments show that Random Forest is effective for wine quality classification. Changing the number of trees and tree depth affects performance. Feature importance analysis shows that alcohol, sulphates, and volatile acidity are among the strongest factors influencing wine quality prediction.

Git Link: <https://github.com/Himasekhar2/Individual-assignment>